
Jiangshan MU

He/Him/His, Postdoctoral Researcher

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Publons ID: <https://www.webofscience.com/wos/author/record/HPE-9830-2023>

Researchgate ID: <https://www.researchgate.net/profile/Jiangshan-Mu-2>

Personal Page: <https://jiangshanmu.github.io/mujsh.github.io/>

Research Interests

- Health and exposure disparities
- Environmental data construction using machine learning
- Urbanization process and global inequalities

Education

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| • | Ph.D. | Shandong University | Environmental Science | 2021.09–2025.06 |
| • | M.S. | Shandong University | Environmental Science | 2018.09–2021.06 |
| • | B.Sc. | Ocean University of China | Environmental Science | 2014.09–2018.06 |

Research Experience

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| • | Postdoctoral Researcher | 2025.10–Present |
| Nicholas School of the Environment, Duke University (Supervisor: Dr. Drew Shindell) | | |
| Research Focus: Investigate the impacts of air pollution and climate variability on human health and socioeconomic systems. | | |
| • | Ph.D. | 2021.09–2025.06 |
| Environment Research Institute, Shandong University (Supervisor: Dr. Likun Xue and Dr. Yuqiang Zhang) | | |
| Research Focus: Construct high-resolution pollutant datasets and explore the disparities in global human health exposure. | | |
| • | Visiting Scholar | 2024.01–2024.05 |
| Industrial Engineering School, Universidad Politécnica de Madrid (Supervisor: Dr. Joaquín B. Ordieres Meré) | | |
| Research Focus: Explore the relationship between extreme events and air pollution based on advanced machine learning. | | |
| • | M.S. | 2018.09–2021.06 |
| Environment Research Institute, Shandong University (Supervisor: Dr. Likun Xue) | | |
| Research Focus: Develop instruments and explore the chemical O ₃ formation regime using | | |

instruments and models.

Professional Skills

- **Machine Learning:** Skilled in using TensorFlow and Scikit-learn for developing predictive models and data analysis, such as: Random Forest, XGBOOST, Transformer, etc.
- **Coding and Modeling:** Proficient in Python and R languages, and experienced with using models such as OBM, PMF, HYSPLIT, etc.
- **Instruments:** Proficient in a range of monitoring instruments, including I-CIMS, WPS, Thermo Scientific Trace Gas Analyzer, etc.

Publications (selected)

Till May. 2026, 39 published articles; First author 4.

1. **Mu, J.**, Tao, C., Zhang, Y., Liu, Z., Zhang, Y., Zhao, N., Luo, B., Zhou, Q., Zhang, Q., Zhang, H., Xue, L. 2026. Improved global daily nitrogen dioxide concentrations from 2005 to 2023 derived using a deep learning approach. *Earth System Science Data*, 18, 2999–3011.
2. **Mu, J.**, Zhang, Y., Zhang, Y., Liu, Z., Tao, C., Luo, B., Xue, L. 2026. Explainable Machine Learning for High-Resolution Modeling of Long-Term Atmospheric Nitrogen and Sulfur Wet Deposition across the United States. *Environment Science and Technology*, 60, 11, 8661–8670.
3. **Mu, J.**, Zhang, Y., Xia, Z., Fan, G., Zhao, M., Sun, X., Liu, Y., Chen, T., Shen, H., Zhang, Z., Zhang, H., Pan, G., Wang, W., Xue, L. 2023. Two-year online measurements of volatile organic compounds (VOCs) at four sites in a Chinese city: Significant impact of petrochemical industry. *Science of The Total Environment*, 858, 159951.
4. **Mu, J.**, Zhu, Y., Shan, Y., Li, H., Chen, T., Wen, L., Qi, Y., Hu, J., Qi, W., Wang, X., Zhou, X., Zhang, Q., Wang, W., Xue, L. 2020. Aircraft measurements of summer atmospheric particle number concentration and size distribution in Northeast China. *Geochemical (CHINA)*, 49, 03.
5. Tan, Y., Zhang, Y., Wang, T., Chen, T., **Mu, J.**, Xue, L. 2024. Dissecting drivers of ozone pollution during the 2022 multi-city lockdowns in China sheds light on future control direction. *Environment Science and Technology*, 58(16): 6988-6997.
6. Sun, Y., Zhu, Y., Qi, Y., Chen, L., **Mu, J.**, Shan, Y., Yang, Y., Nie, Y., Liu, P., Cui, C., Zhang, J., Liu, M., Zhang, L., Wang, Y., Wang, X., Tang, M., Wang, W., Xue, L. 2024. Measurement report: Atmospheric ice nuclei at Changbai Mountain (2623 m asl) in northeastern Asia. *Atmospheric Chemistry and Physics*, 24, 3241–3256.
7. Zhang, Y., Wang, H., Huang, L., Qiao, L., Zhou, M., **Mu, J.**, Wu, C., Zhu, Y., Shen, H., Wang, G., Wang, T., Wang, W., Xue, L. 2023. Double-edged role of VOCs reduction in nitrate formation: Insights from observations during the China International Import Expo 2018. *Environment Science and Technology*, 57, 42, 15979–15989.
8. Zhang, Y., Dai, J., Li, Q., Chen, T., **Mu, J.**, Brasseur, G., Wang, T., Xue, L. 2023. Biogenic

volatile organic compounds enhance ozone production and complicate control efforts: Insights from long-term observations in Hong Kong. *Atmospheric Environment*, 309, 119917.

9. Li, T., Zhang, Q., Peng, Y., Guan, X., Li, L., **Mu, J.**, Wang, X., Yin, X., Wang, Q. 2023. Contributions of various driving factors to air pollution events: Interpretability analysis from Machine learning perspective. *Environment International*, 173, 107861.
10. Zhang, Y., Xue, L., **Mu, J.**, Chen, T., Li, H., Gao, J., Wang, W. 2022. Developing the maximum incremental reactivity for Volatile Organic Compounds in major cities of central-eastern China. *Journal of Geophysical Research: Atmospheres*, 127(22): e2022JD037296.
11. Zhang, Y., Xue, L., Carter, W. P. L., Pei, C., Chen, T., **Mu, J.**, Wang, Y., Zhang, Q., Wang, W. 2021. Development of ozone reactivity scales for volatile organic compounds in a Chinese megacity. *Atmospheric Chemistry and Physics*, 21(14), 11053-11068.
12. Zhang, Y., Xue, L., Li, H., Chen, T., **Mu, J.**, Dong, C., Sun, L., Liu, H., Zhao, Y., Wu, D., Wang, X., Wang, W. 2021. Source apportionment of regional ozone pollution observed at Mount Tai, North China: application of Lagrangian photochemical trajectory model and implications for control policy. *Journal of Geophysical Research: Atmospheres*, 126.6, e2020JD033519.
13. Liu, Y., Shen, H., **Mu, J.**, Li, H., Chen, T., Yang, J., Jiang, Y., Zhu, Y., Meng, H., Dong, C., Wang, W., Xue, L. 2021. Formation of peroxyacetyl nitrate (PAN) and its impact on ozone production in the coastal atmosphere of Qingdao, North China. *Science of the Total Environment*, 778, 146265.

Awards and Honors

•	Excellent Student, Shandong University	2023
•	First-class academic scholarship, Shandong University	2023
•	Excellent Student, Shandong University	2022
•	First-class academic scholarship, Shandong University	2022
•	Graduate Scholarship, Shandong University	2018-2021
•	Second Prize of in National University Mathematical Modeling Competition, Ocean University of China	2015

Conference and Presentation

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1. **Mu, J.**, (Dec, 2025). *American Geophysical Union (AGU) Annual Meeting*. New Orleans, USA.
 2. **Mu, J.**, (Sept, 2025). *Attendant of the 12th International Conference on Air Benefit and Cost and Attainment Assessment & Symposium on Atmospheric Haze Chemistry (ABaCAS 2025)*. Zhejiang, China.
 3. **Mu, J.**, (Aug, 2025). *Chinese Environmental Science Youth Academic Conference*. Changsha, China.

4. **Mu, J.**, (Dec, 2024). Oral: High-Resolution Global NO₂ Dataset (2005-2023) via Air Transformer Model: Decadal Trends and Spatial Insights. *American Geophysical Union (AGU) Annual Meeting*. Washington, DC, USA.
5. **Mu, J.**, (Oct, 2024). Attendant of the 11th International Conference on Air Benefit and Cost and Attainment Assessment & Symposium on Atmospheric Haze Chemistry (ABaCAS 2024). Shanghai, China.
6. **Mu, J.**, (May, 2024). Oral: The Impact of Wildfires on Atmospheric Nitrogen Deposition in the United States. *Climate Change Doctoral Forum*. Nanjing, China.
7. **Mu, J.**, (May, 2024). *Nature conferences: Air Pollution and Climate Change*. Beijing, China.
8. **Mu, J.**, (Apr, 2024). Poster: The Impact of Wildfires on Atmospheric Nitrogen Deposition in the United States: A Multiple Linear Regression-based Analysis. *The General Assembly 2024 of the European Geosciences Union (EGU)*. Vienna, Austria and online.
9. **Mu, J.**, (Dec, 2023). Oral: The Impact of Wildfires on Atmospheric Nitrogen Deposition in the United States. *29th China Conference on Atmospheric Environment Science and Technology and Annual Meeting of Chinese Society for Environmental Sciences - Atmospheric Environment Branch*. Hainan, China and online.
10. **Mu, J.**, (Apr, 2023). *The 4th Academic Forum on Biogeochemical Cycle of Nitrogen*. Beijing, China.
11. **Mu, J.**, (Mar, 2023). *IntelliAQ workshop on Machine Learning for Air Quality*. IntelliAQ Workshop. Cologne, Germany and online.
12. **Mu, J.**, (Nov, 2022). *28th China Conference on Atmospheric Environment Science and Technology and Annual Meeting of Chinese Society for Environmental Sciences - Atmospheric Environment Branch*. Beijing, China and online.
13. **Mu, J.**, (Nov, 2021). Oral: Analysis of atmospheric VOCs pollution characteristics and sources in Ji'nan City. *27th China Conference on Atmospheric Environment Science and Technology and Annual Meeting of Chinese Society for Environmental Sciences - Atmospheric Environment Branch*. Beijing, China and online.

Activities

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| • Committee of the 1st Meteorological Congress | 2025 |
| • Committee of the 10 th International Conference on Air Benefit and Cost and Attainment Assessment & Symposium on Atmospheric Haze Chemistry (ABaCAS 2023) | 2023 |
| • Volunteer of the 24 th Atmospheric Environmental Science and Technology Conference | 2018 |

Journal Reviewer for articles in *Environmental Science & Technology*, *Atmospheric Chemistry and Physics*, *Atmospheric Environment*, *Environmental Pollution*, *Journal of Cleaner Production*, etc.